

Detecting Miscitation on the Scholarly Web through LLM-Augmented Text-Rich Graph Learning

Huidong Wu
Chinese Academy of Sciences &
City University of Hong Kong
Beijing, China
wuhuidong21@mails.ucas.ac.cn

Haojia Xiang
City University of Hong Kong
Hong Kong, China
haojxiang2-c@my.cityu.edu.hk

Jingtong Gao
City University of Hong Kong
Hong Kong, China
jt.g@my.cityu.edu.hk

Xiangyu Zhao*
City University of Hong Kong
Hong Kong, China
xianzhao@cityu.edu.hk

Dengsheng Wu*
Shenzhen University &
Beijing Dongbi Data Technology
Shenzhen, China
wds@szu.edu.cn

Jianping Li
University of Chinese Academy of
Sciences
Beijing, China
ljp@ucas.ac.cn

Abstract

Scholarly web is a vast network of knowledge connected by citations. However, this system is increasingly compromised by miscitation, where references do not support or even contradict the claims they are cited for. Current miscitation detection methods, which primarily rely on semantic similarity or network anomalies, struggle to capture the nuanced relationship between a citation's context and its place in the wider network. While large language models (LLMs) offer powerful capabilities in semantic reasoning for this task, their deployment is hindered by hallucination risks and high computational costs. In this work, we introduce LLM-Augmented Graph Learning-based Miscitation Detector (LAGMiD), a novel framework that leverages LLMs for deep semantic reasoning over citation graphs and distills this knowledge into graph neural networks (GNNs) for efficient and scalable miscitation detection. Specifically, LAGMiD introduces an evidence-chain reasoning mechanism, which uses chain-of-thought prompting, to perform multi-hop citation tracing and assess semantic fidelity. To reduce LLM inference costs, we design a knowledge distillation method aligning GNN embeddings with intermediate LLM reasoning states. A collaborative learning strategy further routes complex cases to the LLM while optimizing the GNN for structure-based generalization. Experiments on three real-world benchmarks show that LAGMiD achieves state-of-the-art miscitation detection with significantly reduced inference cost. The implementation code is available at: https://github.com/Applied-Machine-Learning-Lab/WWW2026_LAGMiD.

CCS Concepts

• Information systems → Web mining.

*Corresponding author



This work is licensed under a Creative Commons Attribution 4.0 International License. WWW '26, Dubai, United Arab Emirates.

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792568>

Keywords

Miscitation Detection, Knowledge Distillation, Graph Neural Networks, Large Language Models, Scholarly Web

ACM Reference Format:

Huidong Wu, Haojia Xiang, Jingtong Gao, Xiangyu Zhao, Dengsheng Wu, and Jianping Li. 2026. Detecting Miscitation on the Scholarly Web through LLM-Augmented Text-Rich Graph Learning. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3774904.3792568>

1 Introduction

The scholarly web constitutes a vast, dynamic, and interconnected ecosystem of knowledge [4, 46, 72], where citation functions as its foundational currency [7, 38]. Through citations, researchers contextualize their work, substantiate claims, and construct upon existing research [58]. The integrity of citation network is, therefore, paramount to the advancement of science. However, this foundation is increasingly undermined by the pervasive issue of miscitation, a phenomenon wherein a referenced source fails to support, or even contradicts, the assertion it is cited to uphold [6, 8, 36, 37]. Empirical estimates indicate that up to 25% of citations in the general scientific literature contain inaccuracies and mislead academics [37]. Whether arising from unintentional oversight or deliberate rhetorical manipulation, miscitations propagate misinformation [8, 20], distort the outputs of academic search engines [54, 74], and ultimately erode the collective trust in the scientific record [2].

The intrinsic graph nature of the scholarly web has motivated a substantial line of research to formulate miscitation detection as an edge classification task [19, 25, 27, 33, 57]. Early methods relied predominantly on network topology, identifying suspicious citations through structural anomalies [19, 57] such as atypical cross-disciplinary linkages (Figure 1(a)). While capable of revealing macro-level relational patterns, these approaches fundamentally overlook the semantic content of the citation context [27], which lies at the heart of citation validity. Subsequent efforts integrated local textual evidence by encoding features from citation sentences to augment classification [27, 42, 47](Figure 1(b)). Despite these improvements, many models still operate on surface-level lexical similarities and lack the depth of semantic understanding required to distinguish strategically inserted or weakly grounded references.

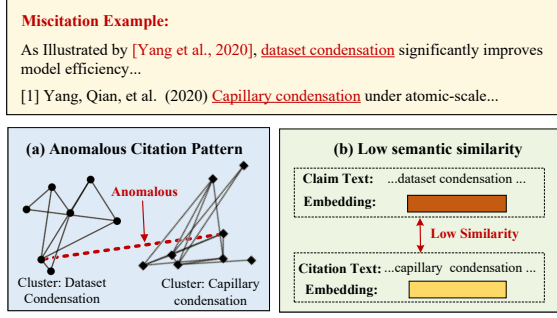


Figure 1: Illustrations of traditional miscitation detection methods. (a) Detection via anomalous citation patterns. (b) Detection via low semantic similarity.

The recent advent of large language models (LLMs) presents a transformative opportunity by offering profound semantic understanding and generative reasoning capabilities [5, 69]. In principle, LLMs can meticulously analyze the citing context against the cited paper’s content, articulating rationales to assess relevance and accuracy [1, 10, 45, 53, 70]. However, this powerful semantic analysis remains inherently local and computationally expensive. First, LLMs are susceptible to hallucinations, particularly when provided with incomplete or biased local context [21, 41]. Since they operate without awareness of the global citation network, they remain blind to systematic manipulation patterns such as when an author distorts the meaning of a reference or fabricates a supporting claim [9, 40]. Such structural anomalies can only be identified through a broader, network-level perspective [17, 21]. These limitations highlight the need for a hybrid architecture that synergizes the deep semantic reasoning of LLMs with the global pattern recognition capabilities of graph models [63]. Second, inference with large language models incurs substantial computational cost. As the volume of scientific publications grows exponentially, so does the number of references requiring analysis [57]. The sheer scale of the scholarly web, with its billions of citation edges, makes fine-grained, context-aware analysis using LLMs computationally intractable for the entire network [15, 22, 32, 43]. These challenges underscore the urgency of developing a robust and scalable method for miscitation detection.

In this study, we propose LAGMiD (LLM-Augmented Graph Learning-based Miscitation Detector), a novel framework that integrates the complementary strengths of LLM-based reasoning and graph-structured learning to identify miscitations across the scholarly web. To mitigate the risk of LLM hallucinations arising from limited local context, we introduce an evidence-chain reasoning mechanism powered by LLMs. This process leverages the inherent knowledge of the LLM to perform chain-of-thought reasoning [56] over text-rich citation graphs by tracing reference sources. In other words, the judgment of a citation must be traced back to multi-hop citation sources. To overcome the substantial computational overhead of applying LLM reasoning at web scale, we propose distilling the LLM’s reasoning capability into a Graph Neural Network (GNN). The evidence-chain reasoning of the LLM aligns naturally with the message-passing mechanism of GNNs. Through a knowledge distillation strategy, we enable the GNN to internalize the LLM’s reasoning patterns, thereby achieving both

faster inference and effective reasoning over graph-structured interaction data. Furthermore, given the diverse nature of miscitations where some manifest as anomalous structural patterns while others require deep semantic context analysis, we design a collaborative learning process that continuously aligns LLM knowledge with GNN structural representations. By iteratively refining GNN learning with LLM knowledge distillation across layers, our approach adaptively captures both structural and semantic anomalies, while simultaneously enhancing operational efficiency.

The main contributions of this paper are summarized as follows:

- (1) We identify key limitations in existing miscitation detection methods and propose LAGMiD. To the best of our knowledge, this is the first miscitation detection framework that integrates LLM reasoning with GNN structural modeling under a unified graph-learning paradigm.
- (2) We design an LLM-based evidence-chain mechanism using chain-of-thought prompting, and transfer this reasoning process to a student GNN via a novel knowledge distillation approach, enabling scalable and efficient inference.
- (3) We introduce a collaborative learning strategy that iteratively aligns GNN representations with distilled LLM knowledge, allowing both models to fully exploit their respective strengths while enhancing overall inference efficiency.
- (4) We perform comprehensive experiments on three real-world datasets, demonstrating that LAGMiD consistently outperforms existing methods in miscitation detection.

2 Preliminary

2.1 Text-Rich Citation Graph

Scholarly webs naturally constitute a category of text-rich graphs (see Appendix A for construction details), where the node set comprises publication nodes and directed edges represent citations that denote referential relationships between nodes [7, 26]. Formally, we define a text-rich citation graph as

$$\mathcal{G} = (\mathcal{P}, \mathcal{E}, \mathcal{X}^{(P)}, \mathcal{X}^{(E)}) \quad (1)$$

where $\mathcal{P}$ denotes the set of publication nodes and $\mathcal{E} \subseteq \mathcal{P} \times \mathcal{P}$ represents the set of directed citation edges. Each node $p \in \mathcal{P}$ is associated with raw textual attributes $\mathcal{X}^{(P)}$, typically including the title, abstract, and/or full body of the publication. Similarly, each edge $e \in \mathcal{E}$ may be associated with textual attributes $\mathcal{X}^{(E)}$, such as citation contexts or explanatory claims extracted from the citing document. The raw texts in the citation graph can be encoded into dense vector representations via language model transformations:

$$\mathbf{X}^{(P)} = f_{\text{LM}}(\mathcal{X}^{(P)}), \quad \mathbf{X}^{(E)} = g_{\text{LM}}(\mathcal{X}^{(E)}) \quad (2)$$

where f_{LM} and g_{LM} denote suitable language model encoders for nodes and edges, respectively.

2.2 Miscitation Detection Problem

Building upon the formulation of a text-rich citation graph, we formalize the task of miscitation detection as a binary edge classification problem. Given a citation graph $\mathcal{G}$ and the annotated edge labels $\mathcal{Y} = \{0, 1\}$, where $y_{ij} = 1$ denotes a miscitation from paper p_i to paper p_j and $y_{ij} = 0$ a valid citation, the goal is to learn a binary classifier f_{θ} that assesses the semantic alignment between

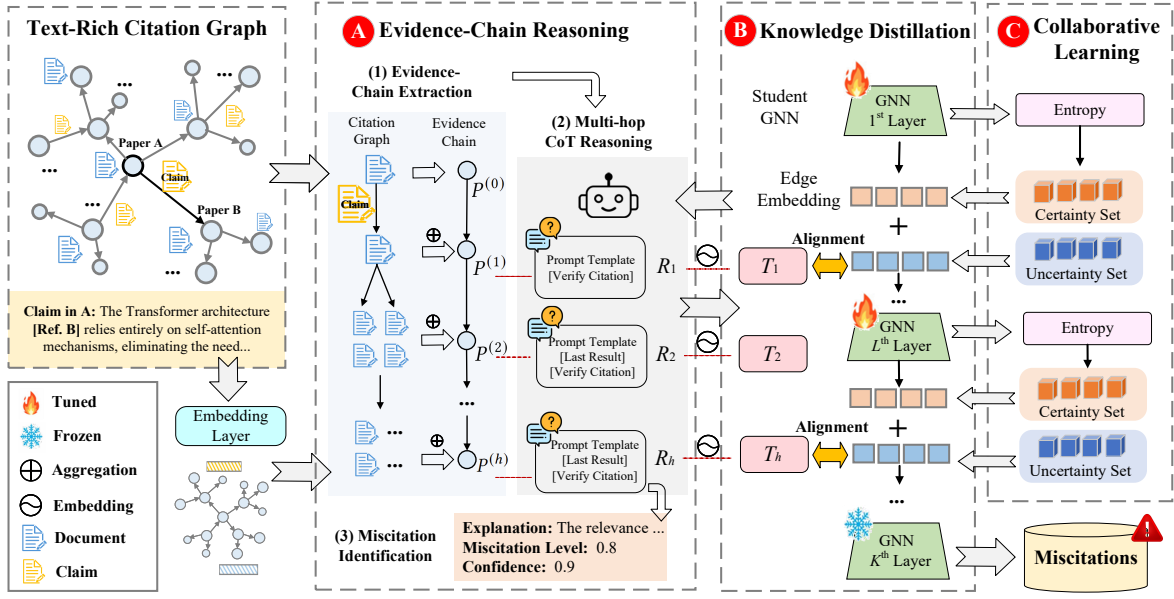


Figure 2: The overview of the proposed LAGMiD framework.

citing statements and cited documents. The model may incorporate multiple sources of evidence, including node features $\mathcal{X}^{(P)}$, edge features $\mathcal{X}^{(E)}$, and the structural properties of the global citation graph $\mathcal{G}$. The classification function is defined as:

$$f_{\theta}(e_{ij}) = \Psi(p_i, p_j, e_{ij}, \mathcal{G}) \rightarrow \{0, 1\} \quad (3)$$

where Ψ is a parameterized function that combines multiple evidence to output a binary prediction. Here, $f_{\theta}(e_{ij}) = 1$ indicates a miscitation, while $f_{\theta}(e_{ij}) = 0$ corresponds to a valid citation.

3 Methodology

In this section, we introduce the proposed LLM-Augmented Graph Learning-based Miscitation Detector (LAGMiD). As illustrated in Figure 2, LAGMiD is designed to leverage the complementary strengths of LLMs and GNNs to enhance the accuracy and robustness of miscitation detection. The architecture of LAGMiD is built around three tightly coupled components. First, it incorporates an LLM-driven evidence-chain reasoning mechanism that enables semantic analysis of citation validity by tracing reference sources through citation graphs (Figure 2A). Second, it employs a knowledge distillation process that transfers the reasoning capabilities of LLMs into a more efficient GNN model, thereby facilitating scalable inference (Figure 2B). Finally, LAGMiD adopts a collaborative learning strategy that jointly optimizes both the LLM and GNN components, enabling mutual enhancement through iterative interaction and shared objectives (Figure 2C). The following subsections provide a detailed explanation of each component within the LAGMiD framework.

3.1 LLM-based Evidence-Chain Reasoning

Although LLMs exhibit a strong capacity for understanding semantic relationships between statements and their citation contexts, they are prone to hallucinations and misjudgments [21, 41]. These

issues may manifest in the form of selective citation, misattribution, or distortion of the original conclusions. Passive reliance on direct statement-citation reasoning is inherently unstable. To address this, we introduce an active reasoning process termed Evidence-Chain Reasoning over the citation graph. The core mechanism of this approach requires the LLM to proactively traverse multi-hop citation chains, thereby validating the integrity of the evidence underlying a given claim. Leveraging the LLM’s chain-of-thought capabilities, our approach performs stepwise semantic reasoning across extracted citation chains.

Step 1: Evidence-Chain Extraction. We begin by extracting a multi-hop evidence chain from the text-rich citation graph $\mathcal{G}$, with the goal of tracing the source documents that support a given claim to enable systematic and verifiable reasoning. For a given citation edge $e_{ij} = (p_i, p_j) \in \mathcal{E}$, where paper p_i cites paper p_j , we treat the statement text $X_{ij}^{(E)}$ as the claim to be assessed. To identify supporting evidence, we construct a source-oriented subgraph rooted at e_{ij} , expanding along citation links for up to K hops. Let $\mathcal{N}^+(p)$ denote the set of direct out-neighbors of node p , representing papers cited by p . The subgraph is generated iteratively as follows:

$$\mathcal{S}_1 = p_j, \quad \mathcal{S}_h = \bigcup_{p \in \mathcal{S}_{h-1}} \mathcal{N}^+(p), \quad \text{for } h = 2, \dots, K \quad (4)$$

where $\mathcal{S}_h$ contains all documents reachable from p_j within h hops over the citation graph.

To ensure relevance and reduce redundancy, we apply a filtering strategy to $\mathcal{S}_h$ based on the vector representations $X^{(P)}$ and $X^{(E)}$ defined in the preliminary section. Let $X_{ij}^{(E)}$ represent the encoded form of the claim context, and $X_u^{(P)}$ the embedding of the textual content of a candidate paper p_u . Using a similarity function $\text{sim}(X_{ij}^{(E)}, X_u^{(P)})$, such as cosine similarity, we select the top- m most semantically relevant nodes $\mathcal{S}'_h$ from $\mathcal{S}_h$. Based on this filtered

candidate set, we construct the evidence chain as a directed path:

$$\pi_{ij} = \left(p^{(0)}, p^{(1)}, \dots, p^{(h)}, \dots, p^{(K)} \right) \quad (5)$$

where $p^{(0)} = p_i$, $p^{(1)} = p_j$ and each $p^{(h)}$ is formed by aggregating the textual content $\mathcal{X}_u^{(P)}$ for all $p_u \in \mathcal{S}'_h$. This evidence chain serves as the basis for the subsequent reasoning procedure.

Step 2: Multi-Hop Chain-of-Thought Reasoning. Given the extracted evidence chains π_{ij} , we employ a structured chain-of-thought reasoning process to perform stepwise verification across the multi-hop evidence trail. The goal of this stage is to determine whether each citation within the chain maintains semantic fidelity to the referenced content, thereby ensuring the consistent and accurate propagation of scholarly information.

At each hop h ($1 \leq h \leq K$), the large language model is prompted to examine the relationship between two consecutive document sets, $p^{(h-1)}$ and $p^{(h)}$. The verification at hop h evaluates whether the citation context $\mathcal{X}_{h-1,h}^{(E)}$ contained in $p^{(h-1)}$ faithfully represents the main claims or conclusions presented in the cited document $p^{(h)}$. Formally, the verification at hop h is defined as:

$$V_h = \text{LLM} \left(\phi_{\text{verify}} \left(\mathcal{X}_{p(h)}^{(P)}, \mathcal{X}_{h-1,h}^{(E)}, \mathcal{X}_{p(h-1)}^{(P)} \right) \right) \quad (6)$$

where ϕ_{verify} denotes a task-specific prompting function that structures the verification input, and V_h represents the verification outcome at hop h , indicating the semantic consistency.

The reasoning process for claim e_{ij} evolves iteratively based on the verification results V_h . Each step integrates newly obtained evidence with the reasoning accumulated from previous hops. The reasoning state at hop h is computed as:

$$R_h = \text{LLM} \left(\phi_{\text{cot}} \left(R_{h-1}, V_h, \mathcal{X}_{p(h)}^{(P)}, \mathcal{X}_{h-1,h}^{(E)} \right) \right) \quad (7)$$

where R_h denotes the updated reasoning state after incorporating the result of hop h , and ϕ_{cot} governs the evolution of the reasoning chain by combining the new verification with the accumulated contextual understanding.

The reasoning process continues sequentially until the terminal node $p^{(K-1)}$ is reached. The complete reasoning trajectory is expressed as $\mathcal{R}_{ij} = [R_1, \dots, R_K]$, which captures the progressive flow of evidence along the chain and documents the semantic integrity of each citation step. This cumulative reasoning trace provides a verifiable foundation for detecting potential miscitation or semantic distortion within the citation network. A detailed case study illustrating this reasoning process is provided in Appendix D.

Step 3: Miscitation Identification. Building upon the reasoning trajectory $\mathcal{R}_{ij} = [R_1, \dots, R_K]$ derived from the multi-hop verification process, the final step produces a structured assessment of the target citation edge e_{ij} . The LLM synthesizes the accumulated reasoning states and verification outcomes to generate a comprehensive miscitation judgment, formatted as a structured JSON object. This output comprises three core components: (1) a natural language explanation U that summarizes the semantic (in)consistencies observed throughout the evidence chain; (2) a miscitation level O , represented as a normalized score in the range $[0, 1]$, quantifying the severity of semantic misalignment, where higher values indicate more significant miscitations; and (3) a confidence score C , also in the range $[0, 1]$, indicating the model's

certainty in its assessment. Formally, the miscitation identification process is defined as:

$$\mathcal{M}_{ij}(U, O, C) = \text{LLM} \left(\phi_{\text{judge}} \left(\mathcal{R}_{ij}, \pi_{ij}, \mathcal{X}_{ij}^{(E)} \right) \right) \quad (8)$$

where ϕ_{judge} denotes a specialized prompt template that constrains the model to generate a JSON-structured output conforming to the specified schema. The resulting object $\mathcal{M}_{ij}$ provides both quantitative scoring and qualitative justification for citation quality, enabling transparent and interpretable miscitation detection that can be directly utilized in downstream analysis pipelines.

3.2 LLM-to-GNN Knowledge Distillation

While LLMs excel at evidence-chain reasoning over citation networks, their application to large-scale scholarly graphs remains constrained by high computational costs [15, 22, 32, 43]. To address this, we propose a knowledge distillation framework that transfers the reasoning capabilities of LLMs to a more efficient graph neural network (GNN). The inherent message-passing mechanism of GNNs offers a natural structural parallel to multi-hop evidence-chain reasoning structure of LLMs [66, 67]. Building on this, we introduce a knowledge distillation method that aligns LLM token embeddings with GNN node representations across hops.

We begin by extracting structured knowledge representations from the evidence-chain reasoning process of the large language model (LLM). For each reasoning hop h ($1 \leq h \leq K$), we obtain the LLM's hidden representations during the verification phase. Specifically, we extract token-level embeddings from the final transformer layer while the LLM processes the chain-of-thought prompt ϕ_{cot} at hop h . Formally, for a reasoning state $\mathcal{R}_h$ at hop h , the LLM generates hidden representations as follows:

$$T_h = \text{Transformer}(\mathcal{R}_h) \quad (9)$$

where $T_h \in \mathbb{R}^{N_l}$, and N_l denotes the sequence length at hop h . These representations capture the semantic understanding and intermediate reasoning state of the LLM at each step of the evidence-chain verification, providing crucial signals for miscitation detection.

Our student model is a graph neural network (GNN) that operates on the original citation graph $\mathcal{G}$, which includes node features $X^{(P)}$ and edge features $X^{(E)}$. We employ a K -layer GNN, where each layer updates node representations via neighborhood aggregation. At layer k , the node representation is updated according to:

$$z_u^{(k)} = \text{UPDATE} \left(z_u^{(k-1)}, \text{AGGREGATE} \left(\{z_v^{(k-1)} : v \in \mathcal{N}^+(u)\} \right) \right) \quad (10)$$

where $z_u^{(0)} = X^{(P)}(p_u)$, and $\mathcal{N}^+(u)$ denotes the set of out-neighbors of node u . For edge-level representations between nodes p_i and p_j , which are central to detecting miscitations, we compute:

$$e_{ij}^{(k)} = \text{EDGE} \left(z_i^{(k)}, z_j^{(k)}, X_{ij}^{(E)} \right) \quad (11)$$

Here, the EDGE function is instantiated as a neural network that combines node representations with corresponding edge features. Notably, the GNN's k -th layer representations are aligned with the k -th hop in the LLM's evidence chain, enabling a layer-wise correspondence between the two models.

To facilitate this alignment, we introduce a progressive distillation framework that encourages the GNN to mimic the LLM's

hop-level reasoning representations. For each reasoning hop h , we minimize the distance between the LLM’s representation T_h and the GNN’s corresponding edge representation $e_{h-1,h}^{(h)}$, which represents the citation edge between consecutive papers. We employ a learning objective based on the InfoNCE loss [55]:

$$\mathcal{L}_{\text{distill}}^{(h)} = -\mathbb{E} \left[\log \frac{\exp(\text{sim}(T_h, e_{h-1,h}^{(h)})/\delta)}{\sum_{j=1}^N \exp(\text{sim}(T_h, e_j^{(h)})/\delta)} \right] \quad (12)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, δ is a temperature hyperparameter, and negative samples are drawn from non-matching edge representations within the same training batch. The overall distillation loss aggregates contributions across all reasoning hops:

$$\mathcal{L}_{\text{distill}} = \sum_{h=1}^K \lambda_h \mathcal{L}_{\text{distill}}^{(h)} \quad (13)$$

Here, λ_h denotes the layer-wise weighting coefficients, which are used to balance the contribution of each hop.

3.3 Iterative Collaborative Learning Strategy

While knowledge distillation from LLMs to GNNs provides a pathway for transferring semantic reasoning capabilities, applying this process uniformly across the entire training set is neither computationally efficient nor optimally effective. GNNs inherently excel at capturing structural dependencies within citation graphs and can resolve a substantial portion of miscitation cases through topological patterns alone [24, 25, 34]. LLMs, conversely, offer superior performance in contextual interpretation and complex semantic reasoning [12, 16, 52, 68]. To leverage these complementary strengths, we design an iterative collaborative learning framework that facilitates selective, targeted distillation.

The iterative collaboration begins with the GNN performing inference over the citation graph to generate node and edge representations. It then identifies regions of high predictive uncertainty, i.e., instances where the GNN itself exhibits low confidence and is likely to benefit from semantic verification of LLMs. For each citation edge e_{ij} , we quantify predictive uncertainty using the entropy of the GNN’s output distribution:

$$U_{ij} = -\sum_{b=1}^B \hat{y}_{ij,b} \cdot \log(\hat{y}_{ij,b}) \quad (14)$$

where $\hat{y}_{ij,b}$ denotes the predicted probability for class b in a binary classification setting ($B = 2$). This entropy metric serves as a reliable indicator of the GNN’s confidence and helps identify candidates for LLM-based refinement. To enable multi-hop alignment, uncertainty is evaluated at each GNN layer $k \in 1, 2, \dots, K$. For each layer k , the top $w\%$ edges with the highest uncertainty values are selected to form a candidate set $\mathcal{E}_{\text{uncertain}}^{(k)}$.

For each uncertain edge $e_{ij} \in \mathcal{E}_{\text{uncertain}}^{(k)}$, the LLM performs evidence-chain reasoning as described in Section 3.1. To ensure the quality of the knowledge being transferred, we apply a filtering mechanism: an edge is included in the distillation set $\mathcal{D}_{\text{distill}}$ only if (1) the LLM’s confidence score C exceeds a predefined threshold τ_{conf} , and (2) the LLM’s prediction aligns with the available ground-truth label. This selective filtering mitigates the risk of

noise propagation and ensures that only high-confidence LLM reasoning guides the GNN’s learning process. Targeted knowledge distillation is applied to the filtered set $\mathcal{D}_{\text{distill}}$. The distillation loss is now specifically computed over this high-quality subset:

$$\mathcal{L}_{\text{distill}}(\mathcal{D}_{\text{distill}}) = \sum_{h=1}^K \lambda_h \mathcal{L}_{\text{distill}}^{(h)}(\mathcal{D}_{\text{distill}}) \quad (15)$$

where $\mathcal{L}_{\text{distill}}^{(h)}(\mathcal{D}_{\text{distill}})$ denotes the knowledge distillation loss at hop h evaluated specifically over the edges in $\mathcal{D}_{\text{distill}}$.

In parallel, the GNN is also optimized for the primary miscitation detection task via a supervised objective. The task-specific loss is defined as the binary cross-entropy over labeled edges:

$$\mathcal{L}_{\text{task}} = -\frac{1}{|\mathcal{E}|} \sum_{e_{ij} \in \mathcal{E}} [y_{ij} \log(\hat{y}_{ij}) + (1 - y_{ij}) \log(1 - \hat{y}_{ij})] \quad (16)$$

where y_{ij} is the ground-truth label for edge e_{ij} , and $\hat{y}_{ij}$ is the GNN’s predicted probability of miscitation. The total training objective combines both losses:

$$\mathcal{L}_{\text{total}} = \beta_1 \mathcal{L}_{\text{distill}}(\mathcal{D}_{\text{distill}}) + \beta_2 \mathcal{L}_{\text{task}} \quad (17)$$

where β_1 and β_2 balance the contributions of targeted knowledge distillation and task-specific supervision.

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets. We evaluate the proposed LAGMiD framework on three real-world scholarly benchmarks: RED (Reference Error Detection) [70], SciFact [47, 48], and S2ORC [31]. The RED dataset is a human-annotated corpus designed for reference error detection, comprising statement–citation pairs extracted from journal articles across diverse domains including biomedical science, chemistry/materials, physics, and social sciences. SciFact serves as a benchmark for scientific claim verification, featuring expert-authored claims paired with relevant abstract sentences, along with corresponding labels and rationale spans. S2ORC constitutes a large-scale English-language academic paper corpus that furnishes comprehensive metadata and citation contexts, making it particularly suitable for miscitation analysis. Following Liu et al. [25], we construct a computer science subset of S2ORC for miscitation detection. The detailed preprocessing of all three datasets is provided in Appendix B.1, and Table 1 reports their statistics.

Table 1: Dataset statistics.

Statistics	RED	SciFact	S2ORC
# Statement-Citation Pairs	250	1,109	3,561
# Miscitations	126	463	936
# Valid Citations	124	646	2,625
# AvgLen	202	244	341
# MaxLen	506	1,085	1,714
# Train	175	809	2,493
# Valid	25	150	356
# Test	50	150	712
# Nodes of Citation Graph	5,844	667	413
# Edges of Citation Graph	5,781	1,559	7,122

Table 2: Overall performance comparison for miscitation detection on the three benchmark datasets. The superior result for each metric is highlighted in bold, and the second-best is underlined. A * denotes cases where the result is statistically significantly better ($p < 0.05$) than the second-best.

Dataset	Metric	GCN	GLAD	RoBERTa	SciBERT	GLM	Qwen	AnomalyLLM	GuARD	LAGMiD (Ours)
RED	AUC	0.7559	0.7774	0.7068	0.7651	0.8220	0.8362	0.8982	<u>0.9100</u>	0.9615*
	F1	0.6857	0.6936	0.6082	0.7471	0.7202	0.8056	0.8354	<u>0.8571</u>	0.9167*
	Precision	0.7557	0.7774	0.5982	0.7182	0.8051	0.7682	0.8327	<u>0.8753</u>	0.9167*
SciFact	AUC	0.6045	0.6244	0.5677	0.6211	0.7909	0.7832	<u>0.8024</u>	0.7906	0.8208*
	F1	0.5138	0.6006	0.5252	0.6622	0.7242	0.7213	<u>0.7958</u>	0.7714	0.8205*
	Precision	0.6673	0.6677	0.6360	0.7102	0.7250	<u>0.7536</u>	0.7485	0.7453	0.7724*
S2ORC	AUC	0.6971	<u>0.7883</u>	0.7459	0.7024	0.7129	0.7506	0.7547	0.7528	0.8100*
	F1	0.6871	<u>0.7932</u>	0.6330	0.7094	0.7621	0.7678	0.7381	0.7687	0.8256*
	Precision	0.6606	0.7733	0.7081	0.7653	0.7681	0.7324	0.7420	<u>0.7740</u>	0.8419*

4.1.2 Evaluation Metrics. For the binary classification task of miscitation detection, we employ three standard evaluation metrics: AUC, F1 score, and Precision. AUC (Area Under the ROC Curve) measures the model’s overall ranking capability across all classification thresholds. Precision specifically quantifies the model’s accuracy in identifying true miscitations among all predicted positives, which is particularly important for minimizing false alarms in practical applications. F1 score, as the harmonic mean of precision and recall, provides a balanced assessment of model performance.

4.1.3 Implementation Details. Our implementation of LAGMiD employs Qwen3-8B as the backbone LLM for evidence-chain reasoning. For the GNN component, we implement a 2-layer GCN architecture with both hidden dimensions of 1024. The edge representation function $\text{EDGE}(\cdot)$ is implemented as a 2-layer MLP that concatenates node representations with edge features. Node and edge textual features are encoded using SciBERT.

We set the evidence chain length $K = 2$ with top- $m = 10$ nodes selected at each hop using cosine similarity filtering. In the collaborative learning framework, we set the confidence threshold $\tau_{\text{conf}} = 0.7$ and select the top $w = 20\%$ most uncertain edges for LLM refinement at each iteration. Other hyperparameters including the distillation temperature δ , layer weights λ_h , and loss weighting coefficients β_1, β_2 are determined through grid search. The model is trained using the AdamW optimizer with an initial learning rate of 0.001 and weight decay of 0.01. We employ a batch size of 32 and train for 200 epochs with early stopping based on validation performance. All experiments are conducted on NVIDIA A100 GPU.

4.1.4 Baselines. We evaluate LAGMiD against eight baseline models spanning four categories: **(1) GNN-based Methods.** This category includes two models: a standard Graph Convolutional Network (GCN) [18], which serves as a fundamental graph-based baseline; and GLAD [27], a GNN that incorporates edge features specifically designed for anomalous citation detection. **(2) Pre-trained Language Model (PLM)-based Methods.** We consider two widely-used PLMs: RoBERTa [30], an optimized Transformer encoder commonly employed as a strong text-based baseline; and SciBERT [3], a domain-specific language model pretrained on scientific corpora. **(3) Large Language Model (LLM)-based Methods.**

This group comprises GLM4-9B [13] and Qwen3-8B [65], two large-scale generative language models capable of capturing complex semantic patterns, making them well-suited for identifying anomalous citations. **(4) Text-Rich Graph Learning Methods.** We also include advanced frameworks designed for text-rich graph learning, which are naturally transferable to miscitation detection. AnomalyLLM [29] leverages LLMs to classify anomalous graph edges by integrating textual and structural information. GuARD [34] performs anomaly detection by aligning rich textual content with graph structure within a unified framework, utilizing a large model as its backbone. Implementation details for all baseline models are provided in Appendix B.2.

4.2 Overall Performance

Table 2 presents the overall performance of our proposed LAGMiD method compared to baseline approaches. Across all evaluated datasets and performance metrics, LAGMiD consistently achieves superior performance. Traditional GNN-based and PLM-based methods, which rely primarily on either structural features or semantic similarity, exhibit the weakest performance. This underscores their limited capability to effectively integrate structural perception with semantic understanding, both of which are critical for accurate miscitation detection. LLM-based methods, which leverage extensive pre-trained knowledge and reasoning capabilities, achieve improved results over traditional models. Nonetheless, they still fall short compared to text-rich graph learning approaches that more effectively integrate graph structural information with semantic reasoning. Among these, our LAGMiD significantly outperforms baselines including AnomalyLLM and GuARD. This performance gap suggests that general-purpose graph anomaly detection techniques may face limitations when adapted to the specific challenges of miscitation detection. Overall, the strong performance of LAGMiD highlights its effectiveness and robustness in addressing the nuanced requirements of miscitation tasks.

4.3 Ablation Study

To rigorously evaluate the contribution of each individual component within the proposed LAGMiD framework, we conduct a comprehensive ablation study. Specifically, we design four model variants, each constructed by removing a particular module:

Table 3: Ablation study on different components of LAGMiD.

Model	RED			S2ORC		
	AUC	F1	Precision	AUC	F1	Precision
w/o EC	0.9295	0.8412	0.8571	0.7480	0.7855	0.7728
w/o KD	0.9487	0.8511	0.8696	0.7586	0.7935	0.7885
w/o LD	0.9503	0.8696	0.8571	0.7700	0.8207	0.8035
w/o TD	0.9487	0.8511	0.8696	0.7625	0.7968	0.8277
LAGMiD	0.9615	0.9167	0.9167	0.8100	0.8256	0.8419

Table 4: Performance comparison of evidence-chain reasoning variants on different foundation models.

LLM	Model	RED			S2ORC		
		AUC	F1	Precision	AUC	F1	Precision
Qwen	Directed	0.8362	0.8056	0.7682	0.7506	0.7678	0.7324
	Unfiltered	0.8486	0.8195	0.7725	0.7429	0.7738	0.7527
	Full	0.8543	0.8333	0.7857	0.7680	0.7855	0.7728
GLM	Directed	0.8220	0.7202	0.8051	0.7129	0.7621	0.7681
	Unfiltered	0.8267	0.7329	0.8100	0.7109	0.6835	0.6654
	Full	0.8385	0.7487	0.8182	0.7504	0.7821	0.7981

- **w/o EC:** Removes evidence chain reasoning, relying solely on direct statement-citation reasoning performed by the LLM.
- **w/o KD:** Replaces knowledge distillation module by a naive feature concatenation strategy.
- **w/o LD:** Removes layer-wise distillation component and performs distillation only at the final representation layer.
- **w/o TD:** Removes targeted distillation strategy based on uncertainty sets, instead applying knowledge distillation uniformly across the entire training dataset.

The results of the ablation study, presented in Table 3, demonstrate that each component of our model contributes significantly to overall performance. Notably, the removal of the evidence chain reasoning module results in a substantial performance drop, underscoring the critical role of multi-hop reasoning in capturing complex citation dependencies. The knowledge distillation mechanism also offers marked improvements over the feature concatenation baseline, emphasizing the effectiveness of aligning the semantic representations between LLMs and GNNs. In addition, the layer-wise distillation strategy consistently enhances performance, suggesting that leveraging reasoning patterns across all GNN layers improves the overall representational quality. Lastly, the targeted distillation approach contributes further performance gains by directing the model’s focus toward more uncertain and informative samples, thereby optimizing the learning process.

4.4 Effectiveness of Evidence-Chain Reasoning

To validate the effectiveness of our evidence-chain reasoning mechanism, we conduct ablation studies on three configurations: (1) **Directed:** without evidence-chain reasoning, using only direct statement-citation analysis; (2) **Unfiltered:** with evidence chains but without semantic filtering; and (3) **Full:** the complete approach with semantic filtering. Experiments are conducted with two LLM backbones (Qwen3-8B and GLM4-9B).

As shown in Table 4, the Full configuration consistently outperforms all other variants. The performance gap between Unfiltered

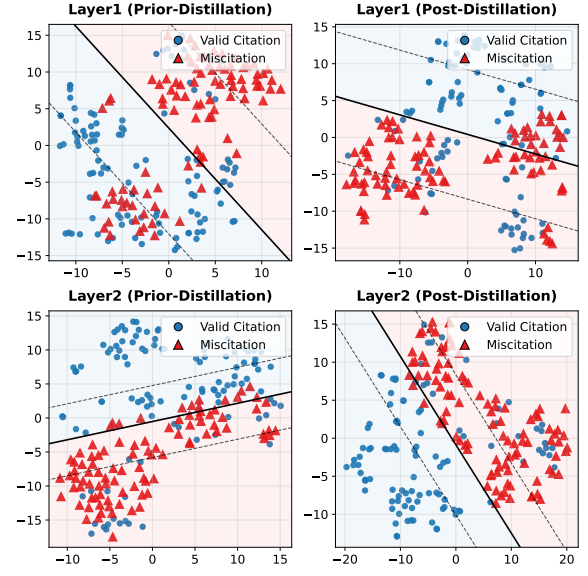


Figure 3: The t-SNE visualizations of citation embeddings from the RED, comparing representations before and after knowledge distillation across different GNN layers. Blue circles denote valid citations, while red triangles represent miscitations. The solid line indicates the decision boundary.

and Directed indicates that simply aggregating all multi-hop evidence does not reliably improve detection. In contrast, the superior performance of Full over Unfiltered underscores the importance of semantic filtering in suppressing noise. These findings collectively demonstrate that multi-hop reasoning, when coupled with semantic filtering, significantly enhances miscitation detection by enabling more accurate and contextually-grounded inference.

4.5 Effectiveness of Knowledge Distillation

We evaluate the effectiveness of our knowledge distillation framework by visualizing embeddings with t-SNE. As illustrated in Figure 3, embeddings from two GNN layers are projected into a two-dimensional space, both before and after distillation. The visualizations reveal that each distillation step enhances the discriminability of GNN embeddings for miscitation detection. Prior to distillation, although some miscitations are identifiable, substantial overlap remains between miscitations and legitimate citations. After distillation, ambiguous instances shift closer to the decision boundary, resulting in improved class separation. The second-layer embeddings also exhibit a noticeable improvement, with a more distinct boundary emerging between the two classes.

These findings further validate the effectiveness of our layer-wise distillation strategy. Specifically, after the first-layer distillation and prior to the second-layer distillation, the GNN leveraging semantic information distilled from the LLM demonstrated enhanced classification performance by transferring structural cues. This indicates that the synergistic integration of the LLM and GNN components within the LAGMiD framework effectively facilitates more discriminative representation learning.

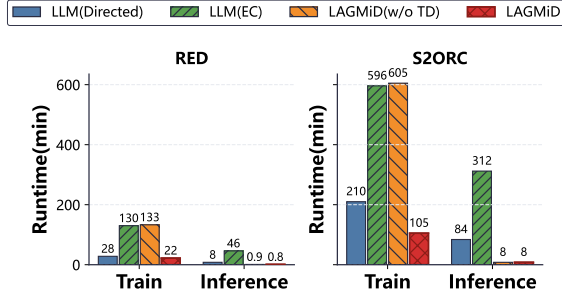


Figure 4: Efficiency comparison of LAGMiD and LLM-based models: Training and inference runtime (in minutes).

4.6 Model Efficiency

To evaluate model efficiency, we compare the runtime of LAGMiD against several LLM-based approaches. As shown in Figure 4, LLM (EC), which employs multi-hop reasoning, incurs significantly higher computational cost than the direct reasoning of LLM (Directed). In contrast, both variants of LAGMiD achieve substantial gains in inference speed. Even without uncertainty filtering, LAGMiD (w/o TD) enables rapid inference by distilling LLM reasoning into the GNN across the full training set. The complete LAGMiD model further improves training efficiency through targeted distillation on the uncertainty subset. The complete LAGMiD model further improves training efficiency through targeted distillation on the uncertainty subset, achieving training times close to those of LLM (Directed). During inference, LAGMiD delivers a 10× speedup over LLM (Directed) and a 100× speedup over LLM (EC), demonstrating strong potential for scalable deployment.

5 Related Work

5.1 Miscitation Detection

Miscitations in scholarly web [8, 20, 44], also referred to as inaccurate or anomalous citations, occur when cited sources do not substantively support the associated statements. Citation-verification studies consistently find that such errors are common, with prevalence estimates 25% in the general scientific literature [37], underscoring the limits of manual peer review and the need for scalable detection. Existing end-to-end detection approaches fall into two main categories. The first leverages semantic similarity by aligning a citing sentence with candidate argument spans in the referenced paper [38, 39, 47, 48]; while this can enforce coarse topical consistency, relatedness does not guarantee semantic adequacy, and fine-grained entailment or reasoning remains challenging. The second mines anomalies in the citation network, for example atypical citation patterns [10, 27, 57] or signals derived from authors, venues, and other metadata [19, 25]. Such methods can reveal strategic or distorted citation behavior, yet they still cannot directly assess whether a claim is supported by its references. Recent LLMs offer a path to fine-grained, semantics-aware miscitation analysis [70], yet web-scale inference is hindered by hallucinations and prohibitive cost. We address this gap by designing an LLM-based reasoning framework for miscitation detection and distilling its knowledge into a GNN. By combining deep semantic understanding with structure-aware modeling of the citation graph, our approach directly evaluates

claim–citation compatibility while retaining the efficiency and scalability required for large-scale deployment.

5.2 Text-Rich Graph Learning

Text-rich graphs, also known as text-attributed graphs, arise when nodes and/or edges carry substantial unstructured text [49, 71]. Classical approaches analyze such graphs by locally encoding text and propagating signals over the topology [59, 60, 75], or by constructing knowledge graphs whose nodes represent words and documents [51]. With the strong text understanding capabilities of LLMs, the integration of LLMs with graphs has become a central direction in text-rich graph learning [16, 23, 73]. Among the LLM-augmented designs most relevant to our setting, three patterns are prominent: LLM-as-Encoder [11, 61], which uses an LLM to produce robust textual embeddings for nodes and edges; LLM-as-Aligner [28, 50, 55], which maps heterogeneous textual and structural signals into a shared space for cross-type reasoning; and retrieval-augmented graph reasoning [14, 35, 62], which constrains or supports LLM judgments using graph neighborhoods. In miscitation detection, textual and structural cues are complementary [27, 64]. Building on existing paradigms, we propose to distilling the LLM’s reasoning process on the graph into a GNN and couple the distilled knowledge with the GNN’s native structural inference. This design achieves more efficient and scalable text-rich graph learning while preserving fine-grained semantic fidelity and exploiting topology-aware generalization.

6 Conclusion

In this paper, we propose LAGMiD, a novel framework that integrates the semantic reasoning power of large language models with graph learning for miscitation detection. By introducing evidence chain reasoning over text-rich citation graphs, distilling such reasoning capacity into a GNN through knowledge transfer, and enabling collaborative optimization with structural features, our method enhances both semantic comprehension of citation contexts and reasoning efficiency. Extensive experiments on three real-world scholarly benchmarks demonstrate that LAGMiD not only achieves state-of-the-art performance in miscitation detection but also significantly accelerates inference compared to LLM-only approaches. Looking forward, we believe the proposed framework holds practical potential in supporting academic integrity initiatives on the scholarly web.

Acknowledgements

This research was supported by grants from the National Natural Science Foundation of China (Nos. T2293774, 72442024, 72022021, 62502404). This research was also partially supported by Hong Kong Research Grants Council (Research Impact Fund No.R1015-23, Collaborative Research Fund No.C1043-24GF, General Research Fund No.11218325), Institute of Digital Medicine of City University of Hong Kong (No.9229503), Huawei (Huawei Innovation Research Program), Tencent (Tencent Rhino-Bird Focused Research Program, Tencent University Cooperation Project), Alibaba (CCF-Alimama Tech Kangaroo Fund No. 2024002), Didi (CCF-Didi Gaia Scholars Research Fund), Kuaishou (CCF-Kuaishou Large Model Explorer Fund, Kuaishou University Cooperation Project), and Bytedance.

References

- [1] Carlos Alvarez, Maxwell Bennett, and Lucy Lu Wang. 2024. Zero-shot scientific claim verification using LLMs and citation text. In *Proceedings of the Fourth Workshop on Scholarly Document Processing (SDP 2024)*. 269–276.
- [2] Xiaomei Bai, Fuli Zhang, Jiaying Liu, Xiaoxia Wang, and Feng Xia. 2025. Revolutionizing scholarly impact: advanced evaluations, predictive models, and future directions. *Artificial Intelligence Review* 58, 10 (2025), 311.
- [3] Iz Beltagy, Kyle Lo, and Arman Cohan. 2019. SciBERT: A Pretrained Language Model for Scientific Text. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. 3615–3620.
- [4] Caroline Birkle, David A Pendlebury, Joshua Schnell, and Jonathan Adams. 2020. Web of Science as a data source for research on scientific and scholarly activity. *Quantitative science studies* 1, 1 (2020), 363–376.
- [5] Daniil A Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. 2023. Autonomous chemical research with large language models. *Nature* 624, 7992 (2023), 570–578.
- [6] Lutz Bornmann and Christian Leibel. 2025. Citation accuracy, citation noise, and citation bias: A foundation of citation analysis. *arXiv preprint arXiv:2508.12735* (2025).
- [7] Qiguang Chen, Mingda Yang, Libo Qin, Jinhao Liu, Zheng Yan, Jiannan Guan, Dengyun Peng, Yiyang Ji, Hanjing Li, Mengkang Hu, et al. 2025. AI4Research: A Survey of Artificial Intelligence for Scientific Research. *arXiv preprint arXiv:2507.01903* (2025).
- [8] Cory L Cobb, Brianna Crumly, Pablo Montero-Zamora, Seth J Schwartz, and Charles R Martinez Jr. 2024. The problem of miscitation in psychological science: Righting the ship. *American Psychologist* 79, 2 (2024), 299.
- [9] Xingyu Deng, Xi Wang, and Mark Stevenson. 2025. The next phase of scientific fact-checking: advanced evidence retrieval from complex structured academic papers. In *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR)*. 436–448.
- [10] Farhaan Ebadulla, BV Gaurav, H Manoj, Arti Arya, Nazmin Begum, and K Kruthik. 2025. Detecting Anomalous Self-citations Using Citation Network Analysis and LLMs. In *International Conference on Engineering Applications of Neural Networks*. Springer, 87–99.
- [11] Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2024. Talk like a Graph: Encoding Graphs for Large Language Models. In *The Twelfth International Conference on Learning Representations*. 1–26.
- [12] Zichuan Fu, Xiangyang Li, Chuhan Wu, Yichao Wang, Kuicai Dong, Xiangyu Zhao, Mengchen Zhao, Huifeng Guo, and Ruiming Tang. 2023. A unified framework for multi-domain ctr prediction via large language models. *ACM Transactions on Information Systems* (2023).
- [13] Team GLM, Aohan Zeng, Bin Xu, Bowen Wang, Chenhui Zhang, Da Yin, Dan Zhang, Diego Rojas, Guanyu Feng, Hanlin Zhao, et al. 2024. Chatglm: A family of large language models from glm-130b to glm-4 all tools. *arXiv preprint arXiv:2406.12793* (2024).
- [14] Xiaoxin He, Yijun Tian, Yifei Sun, Nitesh Chawla, Thomas Laurent, Yann LeCun, Xavier Bresson, and Bryan Hooi. 2024. G-retriever: Retrieval-augmented generation for textual graph understanding and question answering. *Advances in Neural Information Processing Systems* 37 (2024), 132876–132907.
- [15] Shengxiang Hu, Guobing Zou, Song Yang, Shiyi Lin, Yanglan Gan, Bofeng Zhang, and Yixin Chen. 2025. Large language model meets graph neural network in knowledge distillation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 17295–17304.
- [16] Bowen Jin, Gang Liu, Chi Han, Meng Jiang, Heng Ji, and Jiawei Han. 2024. Large language models on graphs: A comprehensive survey. *IEEE Transactions on Knowledge and Data Engineering* (2024).
- [17] Bowen Jin, Chulin Xie, Jiawei Zhang, Kashob Kumar Roy, Yu Zhang, Zheng Li, Ruirui Li, Xianfeng Tang, Suhang Wang, Yu Meng, et al. 2024. Graph Chain-of-Thought: Augmenting Large Language Models by Reasoning on Graphs. In *Findings of the 62nd Annual Meeting of the Association for Computational Linguistics, ACL 2024*. Association for Computational Linguistics (ACL), 163–184.
- [18] TN Kipf. 2016. Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907* (2016).
- [19] Sadamori Kojaku, Giacomo Livan, and Naoki Masuda. 2021. Detecting anomalous citation groups in journal networks. *Scientific Reports* 11, 1 (2021), 14524.
- [20] Janne S Kotiaho. 1999. Papers vanish in mis-citation black hole. *Nature* 398, 6722 (1999), 19–19.
- [21] Ernest Lavrinovics, Russa Biswas, Johannes Bjerva, and Katja Hose. 2025. Knowledge graphs, large language models, and hallucinations: An nlp perspective. *Journal of Web Semantics* 85 (2025), 100844.
- [22] Quan Li, Tianxiang Zhao, Lingwei Chen, Junjie Xu, and Suhang Wang. 2024. Enhancing graph neural networks with limited labeled data by actively distilling knowledge from large language models. In *2024 IEEE International Conference on Big Data (BigData)*. IEEE, 741–746.
- [23] Yuhuan Li, Zhixun Li, Peisong Wang, Jia Li, Xiangguo Sun, Hong Cheng, and Jeffrey Xu Yu. 2024. A survey of graph meets large language model: progress and future directions. In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence*. 8123–8131.
- [24] Xixun Lin, Wenxiao Zhang, Fengzhao Shi, Chuan Zhou, Lixin Zou, Xiangyu Zhao, Dawei Yin, Shirui Pan, and Yanan Cao. 2024. Graph neural stochastic diffusion for estimating uncertainty in node classification. In *41st International Conference on Machine Learning (PMLR)*. MLResearchPress.
- [25] Jiaying Liu, Xiaomei Bai, Mengying Wang, Suppawong Tuarob, and Feng Xia. 2024. Anomalous citations detection in academic networks. *Artificial Intelligence Review* 57, 4 (2024), 103.
- [26] Jiaying Liu, Jing Ren, Wenqing Zheng, Lianhua Chi, Ivan Lee, and Feng Xia. 2020. Web of scholars: A scholar knowledge graph. In *Proceedings of the 43rd international ACM SIGIR conference on research and development in information retrieval*. 2153–2156.
- [27] Jiaying Liu, Feng Xia, Xu Feng, Jing Ren, and Huan Liu. 2022. Deep graph learning for anomalous citation detection. *IEEE Transactions on Neural Networks and Learning Systems* 33, 6 (2022), 2543–2557.
- [28] Qidong Liu, Xian Wu, Xiangyu Zhao, Yuanshao Zhu, Zijian Zhang, Feng Tian, and Yefeng Zheng. 2024. Large language model distilling medication recommendation model. *arXiv preprint arXiv:2402.02803* (2024).
- [29] Shuo Liu, Di Yao, Lanting Fang, Zhetao Li, Wenbin Li, Kaiyu Feng, Xiaowen Ji, and Jingping Bi. 2024. Anomalyllm: Few-shot anomaly edge detection for dynamic graphs using large language models. In *2024 IEEE International Conference on Data Mining (ICDM)*. IEEE, 785–790.
- [30] Yinhao Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692* (2019).
- [31] Kyle Lo, Lucy Lu Wang, Mark Neumann, Rodney Kinney, and Daniel S Weld. 2020. S2ORC: The Semantic Scholar Open Research Corpus. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. 4969–4983.
- [32] Bo Pan, Zheng Zhang, Yifei Zhang, Yuntong Hu, and Liang Zhao. 2024. Distilling large language models for text-attributed graph learning. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*. 1836–1845.
- [33] Guansong Pang, Charu Aggarwal, Chunhua Shen, and Nicu Sebe. 2022. Editorial deep learning for anomaly detection. *IEEE Transactions on Neural Networks and Learning Systems* 33, 6 (2022), 2282–2286.
- [34] Yunhe Pang, Bo Chen, Fanjin Zhang, Yanghui Rao, Evgeny Kharlamov, and Jie Tang. 2025. GuARD: Effective Anomaly Detection through a Text-Rich and Graph-Informed Language Model. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*. 2222–2233.
- [35] Boci Peng, Yun Zhu, Yongchao Liu, Xiaohu Bo, Haizhou Shi, Chuntao Hong, Yan Zhang, and Siliang Tang. 2024. Graph retrieval-augmented generation: A survey. *arXiv preprint arXiv:2408.08921* (2024).
- [36] Nicholas Peoples. 2024. Defensive scholarship: learning from academia’s plagiarism crisis. *The Lancet* 403, 10442 (2024), 2374–2375.
- [37] Nicholas Peoples, Truls Østbye, and Lijing L Yan. 2023. Burden of proof: combating inaccurate citation in biomedical literature. *bmj* 383 (2023).
- [38] Fabio Petroni, Samuel Broscheit, Aleksandra Piktus, Patrick Lewis, Gautier Izacard, Lucas Hosseini, Jane Dwivedi-Yu, Maria Lomeli, Timo Schick, Michele Bevilacqua, et al. 2023. Improving Wikipedia verifiability with AI. *Nature Machine Intelligence* 5, 10 (2023), 1142–1148.
- [39] Tiziano Piccardi, Miriam Redi, Giovanni Colavizza, and Robert West. 2020. Quantifying engagement with citations on Wikipedia. In *Proceedings of the web conference 2020*. 2365–2376.
- [40] Ori Press, Andreas Hochlehnert, Ameya Prabhu, Vishal Udandara, Ofir Press, and Matthias Bethge. 2024. CiteME: Can Language Models Accurately Cite Scientific Claims? *Advances in Neural Information Processing Systems* 37 (2024), 7847–7877.
- [41] Abhilasha Ravichander, Shruti Ghela, David Wadden, and Yejin Choi. 2025. Halogen: Fantastic llm hallucinations and where to find them. *arXiv preprint arXiv:2501.08292* (2025).
- [42] Jing Ren, Feng Xia, Ivan Lee, Azadeh Noori Hoshyar, and Charu Aggarwal. 2023. Graph learning for anomaly analytics: Algorithms, applications, and challenges. *ACM Transactions on Intelligent Systems and Technology* 14, 2 (2023), 1–29.
- [43] Amit Roy, Ning Yan, and Masood Mortazavi. 2025. Llm-driven knowledge distillation for dynamic text-attributed graphs. *arXiv preprint arXiv:2502.10914* (2025).
- [44] Michael Schlichtkrull, Zhijiang Guo, and Andreas Vlachos. 2023. Averitec: A dataset for real-world claim verification with evidence from the web. *Advances in Neural Information Processing Systems* 36 (2023), 65128–65167.
- [45] Farzad Shami, Stefano Marchesin, and Gianmaria Silvello. 2025. Fact verification in knowledge graphs using LLMs. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 3985–3989.
- [46] Liwen Vaughan and Mike Thelwall. 2003. Scholarly use of the Web: What are the key indicators of links to journal Web sites? *Journal of the American Society for Information Science and Technology* 54, 1 (2003), 29–38.

- [47] David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu Wang, Madeleine van Zuylen, Arman Cohan, and Hannaneh Hajishirzi. 2020. Fact or Fiction: Verifying Scientific Claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 7534–7550.
- [48] David Wadden, Kyle Lo, Bailey Kuehl, Arman Cohan, Iz Beltagy, Lucy Lu Wang, and Hannaneh Hajishirzi. 2022. SciFact-Open: Towards open-domain scientific claim verification. In *Findings of the Association for Computational Linguistics: EMNLP 2022*. 4719–4734.
- [49] Danny Wang, Ruihong Qiu, Guangdong Bai, and Zi Huang. 2025. Text meets topology: Rethinking out-of-distribution detection in text-rich networks. *arXiv preprint arXiv:2508.17690* (2025).
- [50] Duo Wang, Yuan Zuo, Fengzhi Li, and Junjie Wu. 2024. Llm as zero-shot graph learners: Alignment of gnn representations with llm token embeddings. *Advances in Neural Information Processing Systems* 37 (2024), 5950–5973.
- [51] Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. 2023. Can language models solve graph problems in natural language? *Advances in Neural Information Processing Systems* 36 (2023), 30840–30861.
- [52] Hanbing Wang, Xiaorui Liu, Wenqi Fan, Xiangyu Zhao, Venkataramana Kini, Devendra Pratap Yadav, Fei Wang, Zhen Wen, and Hui Liu. 2025. Rethinking large language model architectures for sequential recommendations. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*. 3376–3391.
- [53] Siyuan Wang, James R Foulds, Md Osman Gani, and Shimei Pan. 2025. LLM-based Corroborating and Refuting Evidence Retrieval for Scientific Claim Verification. *arXiv preprint arXiv:2503.07937* (2025).
- [54] Shuyao Wang, Zhi Zheng, Yongduo Sui, and Hui Xiong. 2025. Unleashing the Power of Large Language Model for Denoising Recommendation. In *Proceedings of the ACM on Web Conference 2025*. 252–263.
- [55] Zixu Wang, Bingbing Xu, Yige Yuan, Huawei Shen, and Xueqi Cheng. 2025. InfoNCE is a Free Lunch for Semantically guided Graph Contrastive Learning. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 719–728.
- [56] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* 35 (2022), 24824–24837.
- [57] Jonathan D Wren and Constantin Georgescu. 2022. Detecting anomalous referencing patterns in PubMed papers suggestive of author-centric reference list manipulation. *Scientometrics* 127, 10 (2022), 5753–5771.
- [58] Dengsheng Wu, Huidong Wu, and Jianping Li. 2024. Citation advantage of positive words: predictability, temporal evolution, and universality in varied quality journals. *Scientometrics* 129, 7 (2024), 4275–4293.
- [59] Dengsheng Wu, Huidong Wu, and Jianping Li. 2024. Hierarchy-aware adaptive graph neural network. *IEEE Transactions on Knowledge and Data Engineering* (2024).
- [60] Dengsheng Wu, Huidong Wu, Fan Meng, and Jianping Li. 2025. Classifying ultra-short scientific texts using a hybrid hierarchical multi-label classification framework. *Journal of the Association for Information Science and Technology* (2025).
- [61] Xixi Wu, Yifei Shen, Caihua Shan, Kaitao Song, Siwei Wang, Bohang Zhang, Jiarui Feng, Hong Cheng, Wei Chen, Yun Xiong, et al. 2024. Can graph learning improve planning in LLM-based agents? *Advances in Neural Information Processing Systems* 37 (2024), 5338–5383.
- [62] Derong Xu, Xinhang Li, Ziheng Zhang, Zhenxi Lin, Zhihong Zhu, Zhi Zheng, Xian Wu, Xiangyu Zhao, Tong Xu, and Enhong Chen. 2025. Harnessing large language models for knowledge graph question answering via adaptive multi-aspect retrieval-augmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 25570–25578.
- [63] Derong Xu, Ziheng Zhang, Zhenxi Lin, Xian Wu, Zhihong Zhu, Tong Xu, Xiangyu Zhao, Yefeng Zheng, and Enhong Chen. 2024. Multi-perspective Improvement of Knowledge Graph Completion with Large Language Models. In *2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. European Language Resources Association (ELRA), 11956–11968.
- [64] Junjie Xu, Zongyu Wu, Minhua Lin, Xiang Zhang, and Suhang Wang. 2024. Llm and gnn are complementary: Distilling llm for multimodal graph learning. *arXiv preprint arXiv:2406.01032* (2024).
- [65] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388* (2025).
- [66] Haoran Yang, Xiangyu Zhao, Muyang Li, Hongxu Chen, and Guandong Xu. 2023. Mitigating the performance sacrifice in DP-satisfied federated settings through graph contrastive learning. *Information Sciences* 648 (2023), 119552.
- [67] Haoran Yang, Xiangyu Zhao, Yicong Li, Hongxu Chen, and Guandong Xu. 2023. An empirical study towards prompt-tuning for graph contrastive pre-training in recommendations. *Advances in neural information processing systems* 36 (2023), 62853–62868.
- [68] Chao Zhang, Haoxin Zhang, Shiwei Wu, Di Wu, Tong Xu, Xiangyu Zhao, Yan Gao, Yao Hu, and Enhong Chen. 2025. NoteLLM-2: Multimodal Large Representation Models for Recommendation. In *31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025)*. Association for Computing Machinery, 2815–2826.
- [69] Qiang Zhang, Keyan Ding, Tianwen Lv, Xinda Wang, Qingyu Yin, Yiwen Zhang, Jing Yu, Yuhao Wang, Xiaotong Li, Zhuoyi Xiang, et al. 2025. Scientific large language models: A survey on biological & chemical domains. *Comput. Surveys* 57, 6 (2025), 1–38.
- [70] Tianmai M Zhang and Neil F Abernethy. 2024. Detecting reference errors in scientific literature with large language models. *arXiv preprint arXiv:2411.06101* (2024).
- [71] Xinyang Zhang, Chenwei Zhang, Xin Luna Dong, Jingbo Shang, and Jiawei Han. 2021. Minimally-supervised structure-rich text categorization via learning on text-rich networks. In *Proceedings of the Web Conference 2021*. 3258–3268.
- [72] Yu Zhang, Yanzhen Shen, SeongKu Kang, Xiusi Chen, Bowen Jin, and Jiawei Han. 2025. Chain-of-factors paper-reviewer matching. In *Proceedings of the ACM on Web Conference 2025*. 1901–1910.
- [73] Jiajun Zhou, Chenxuan Xie, Shengbo Gong, Zhenyu Wen, Xiangyu Zhao, Qi Xuan, and Xiaoni Yang. 2025. Data augmentation on graphs: A technical survey. *Comput. Surveys* 57, 11 (2025), 1–34.
- [74] Caleb Ziems, William Held, Jane Dwivedi-Yu, and Diyi Yang. 2024. Measuring and Addressing Indexical Bias in Information Retrieval. In *Findings of the Association for Computational Linguistics ACL 2024*. 12860–12877.
- [75] Tao Zou, Le Yu, Yifei Huang, Leilei Sun, and Bowen Du. 2023. Pretraining Language Models with Text-Attributed Heterogeneous Graphs. In *Findings of the Association for Computational Linguistics: EMNLP 2023*. 10316–10333.

A Construction of Text-Rich Citation Graph from Scholarly Web

A systematic preprocessing pipeline is required to transform scholarly web data into a text-rich citation graph $\mathcal{G} = (\mathcal{P}, \mathcal{E}, \mathcal{X}^{(P)}, \mathcal{X}^{(E)})$. This process typically begins with data acquisition from diverse scholarly sources, including academic databases such as Web of Science, Semantic Scholar, and PubMed Central. The nodes $\mathcal{P}$ generally represent individual documents, such as scientific papers or Wikipedia entries. Their corresponding raw text attributes $\mathcal{X}^{(P)}$ can be defined at multiple levels of granularity, including titles, abstracts, and section contents. Edges $\mathcal{E}$ in a text-rich citation graph denote explicit citation relationships between publications. Each directed edge $e_{ij} = (p_i, p_j)$ represents a specific citation where paper p_i references paper p_j . Crucially, these edges are also associated with raw text attributes $\mathcal{X}^{(E)}$, which capture the semantic context of the citation relationship. This is achieved by extracting and processing relevant textual information from the citing documents, such as sentences or paragraphs that mention the cited work, and employing advanced natural language processing techniques to identify specific claims or statements about the referenced publication. Such a text-rich citation graph reflects an integrated and interactive characterization of the scholarly web, seamlessly combining structured citation patterns with rich semantic content.

B More Implementation Details

B.1 Dataset Construction and Processing

RED Dataset [70]. This dataset was originally curated for citation-level fact checking, where each instance is a human-annotated claim–citation pair indicating whether the cited source substantiates the claim or constitutes a reference error. Building on this resource, we augment RED with citation-network structure to enable graph-based learning. Specifically, using Digital Object Identifiers (DOIs) as anchors, we attach bibliographic metadata from Web of Science (WoS) and expand the local citation neighborhood

Table 5: Performance analysis of evidence chain length K .

Dataset	AUC			
	$K = 1$	$K = 2$	$K = 3$	$K = 4$
RED	0.8362	0.8543	0.8470	0.8215
S2ORC	0.7206	0.7680	0.7450	0.7312

Table 6: Performance analysis of evidence chain selection parameter m (using Qwen3-8B).

Dataset	AUC			
	$m = 1$	$m = 5$	$m = 10$	$m = 20$
RED	0.8186	0.8543	0.8560	0.8420
S2ORC	0.7200	0.7480	0.7590	0.7460

Table 7: Performance analysis of uncertainty set ratio $w\%$.

Dataset	Metric	Uncertainty Set Ratio $w\%$					
		1%	5%	10%	20%	40%	50%
RED	AUC	0.8600	0.9015	0.9295	0.9615	0.9503	0.9455
	F1	0.8333	0.8511	0.8511	0.9167	0.9020	0.9412
	Precision	0.8276	0.8696	0.8889	0.9167	0.8519	0.8889
S2ORC	AUC	0.7429	0.7712	0.7953	0.8100	0.8037	0.7989
	F1	0.8000	0.7879	0.8224	0.8256	0.8302	0.8211
	Precision	0.7480	0.7620	0.7787	0.8419	0.7700	0.7827

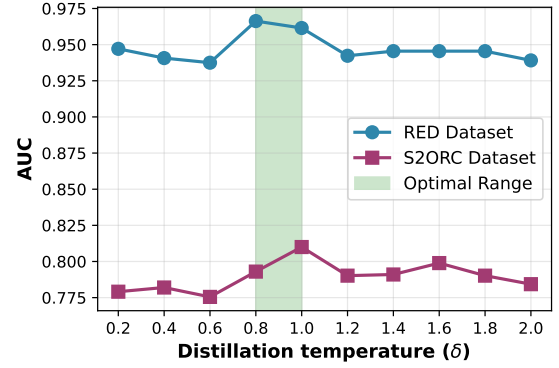
to third-order references, yielding a connected citation graph while preserving the original human-provided labels. For all experiments, we adhere to this augmented RED corpus and partition it into train/dev/test splits in a 7:1:2 ratio to maintain class balance.

SciFact Dataset [47]. This is a scientific-claim verification dataset in which natural-language claims are paired with evidence from peer-reviewed paper abstracts. Each claim is annotated with a veracity label. In our experiments, we follow the official SciFact data setup: we use the dataset’s prescribed train/dev/test splits without modification, and adopt its label space. This alignment ensures direct comparability with prior work.

S2ORC Dataset [31]. To address the scarcity of publicly available datasets for miscitation detection, we construct a computer science subset from S2ORC by following the methodology outlined in [25]. We begin with papers containing complete metadata to build a citation-preserving corpus that maintains authentic references and explicit citing–cited relationships. To simulate miscitation behaviors, we inject synthetically generated anomalous citations into a randomly sampled 50% of the papers, leaving the remainder unaltered. For each sampled paper, the number of injected anomalies is kept equal to that of its original references, maintaining a 1:1 ratio. The injected anomalies cover three distinct types: (i) citations to collaborators’ publications, (ii) within-journal self-promotional citations, and (iii) cross-domain citations that are semantically unrelated to the paper’s content. The corpus is partitioned into training, validation, and test sets in a 7:1:2 ratio.

B.2 Baseline Implementation Details

We benchmark eight models across four families: GNN based, PLM based, LLM based, and text rich graph learning (TGL). To ensure comparability, we use a unified preprocessing and evaluation pipeline.


Figure 5: Performance analysis of distillation temperature δ across datasets.

GNN based Methods. This category includes GCN and GLAD. Both models use a two layer architecture with a hidden size of 1024, ReLU activations, and dropout of 0.5. We train for 200 epochs with Adam (learning rate 1×10^{-3}) and use the same graph structure as well as identical node and edge features across methods.

PLM based Methods. We evaluate RoBERTa and SciBERT. For each instance, we concatenate the claim and the corresponding citation text with a [SEP] delimiter and feed the sequence to the encoder [38, 47]. We use a maximum sequence length of 512, a batch size of 16, and a linearly decayed learning rate of 2×10^{-5} with AdamW. Inputs are tokenized with the model specific tokenizer.

LLM based Methods. We consider two open source LLMs, namely GLM-4-9B and Qwen-3-8B, accessed through their official APIs. We standardize decoding with temperature equal to 0.1 and the default top_p in order to reduce stochasticity. Inputs follow the same claim–citation concatenation scheme with an explicit instruction template for classification. We map generated outputs to discrete labels using fixed regular expressions.

TGL based Methods. From text rich graph learning, we adopt AnomalyLLM and GuARD. Both models operate on text attributed graphs for anomaly detection at the edge level. We treat anomalous edges as miscitations and train edge level detectors accordingly. Following the original recommendations, we construct text features from the claim and citation content using the same PLM encoder across baselines, retain identical graph connectivity, and use the default losses and regularizers for graph based anomaly detection.

C Parameter Analysis

Our approach incorporates several hyperparameters, and we present a detailed analysis of four key ones.

Evidence Chain Length Parameter K . The chain-of-thought length K determines the number of reasoning hops. As shown in Table 5, when $K = 2$, the model achieves the best performance on both the RED and S2ORC datasets. When $K = 1$, the model’s performance is limited due to its inability to leverage multi-hop dependencies; when $K > 2$, excessively long reasoning chains introduce noise and increase computational overhead, leading to performance degradation. Therefore, this paper sets $K = 2$ to achieve the

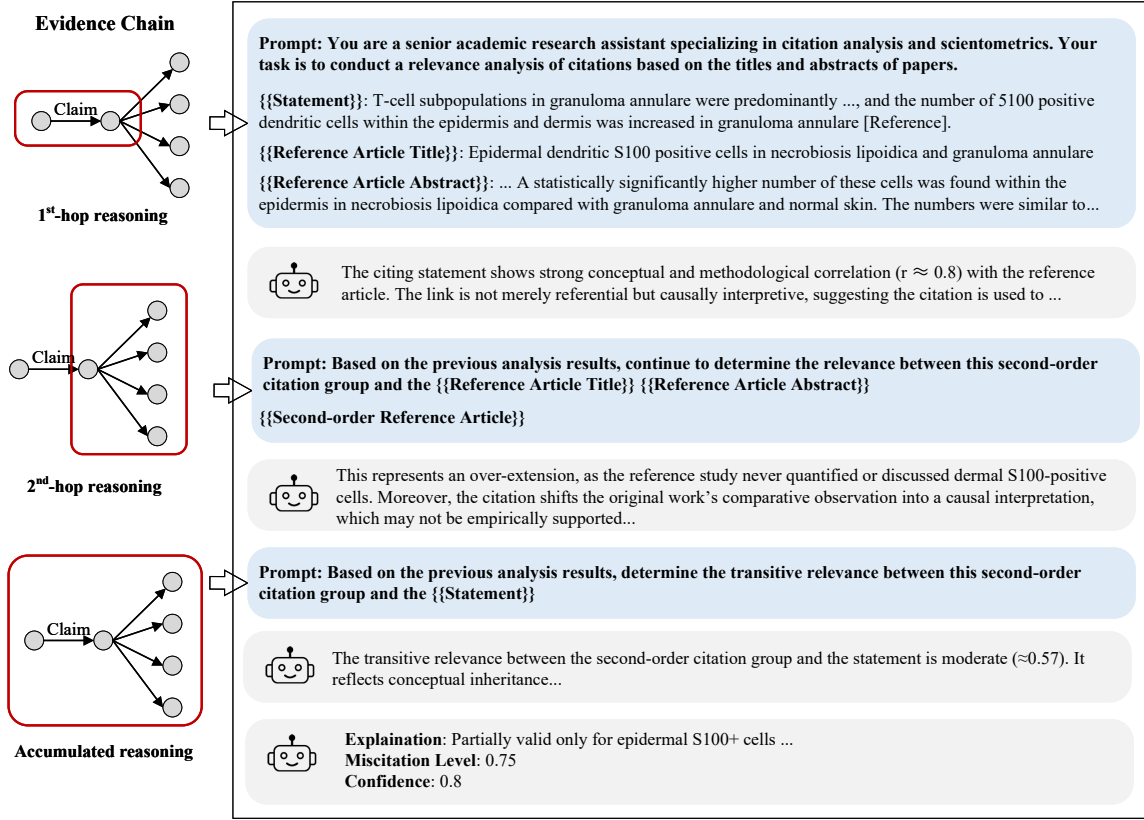


Figure 6: Case study of a 2-hop evidence-chain reasoning process with Chain-of-Thought.

optimal balance between capturing complex citation relationships and maintaining reasoning efficiency.

Evidence Chain Selection Parameter m . The parameter m determines the number of top relevant nodes retained at each hop during evidence chain extraction. As summarized in Table 6, both datasets exhibit similar trends: performance initially improves with larger m , peaks at moderate values ($m = 5-10$), and slightly declines when m becomes too large ($m = 20$). This pattern suggests that selecting too few nodes ($m = 1$) may fail to capture valuable citation sources across multiple hops, while an excessively broad selection ($m = 20$) introduces noise that undermines model performance.

Uncertainty Set Ratio $w\%$. The uncertainty set ratio $w\%$ specifies the proportion of uncertain edges selected for refinement by a large language model (LLM) in our collaborative learning framework. Table 7 provides comprehensive results across different ratios. We observe that both datasets achieve optimal performance at $w\% = 20\%$, with RED attaining a peak AUC of 0.9615 and S2ORC reaching 0.8100. This optimal value reflects a trade-off between computational efficiency and reasoning quality: smaller ratios ($w\% < 20\%$) risk overlooking informative uncertain instances, while larger ratios ($w\% > 20\%$) incur additional computational cost without yielding meaningful performance improvements.

Distillation Temperature Parameter δ . The temperature parameter δ in the InfoNCE loss controls the concentration level of

the knowledge distribution. As illustrated in Figure 5, model performance remains robust across a range of δ values. The optimal range of $\delta = 0.8-1.0$ indicates that moderate temperature settings strike an effective balance between exploration and exploitation in the knowledge distillation process. Excessively low temperatures ($\delta < 0.8$) may produce over-confident distributions that impede knowledge transfer, while overly high values ($\delta > 1.0$) yield excessively smooth distributions.

D Case Study

Figure 6 provides a specific example of our LLM-based evidence-chain reasoning. This case showcases a 2-hop chain-of-thought reasoning procedure. The left side of the figure depicts a schematic of the extracted evidence chain, with red boxes highlighting the scope of reasoning focus at each step. The right side details the corresponding prompt template and the intermediate reasoning outputs generated by the LLM. The process begins with the LLM analyzing the semantic alignment between the initial claim and its direct citation. Subsequently, the model evaluates the validity of the citation in relation to its own 2-hop citation sources. Finally, by synthesizing the multi-step reasoning trajectory, the LLM assesses the overall logical coherence of the claim within the citation chain and determines the degree of miscitation.